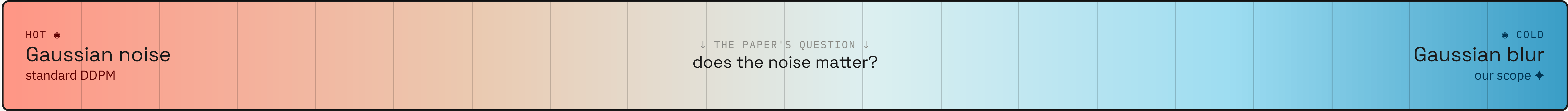


# Cold Diffusion

do diffusion models actually need noise?

Nicholas Josephson · Julius Samwer · Kevin Rodriguez · Tudor Braicu  
Cornell University · CS 4782 Deep Learning · Spring 2026

RE-IMPLEMENTATION  
Bansal et al. · arXiv:2208.09392

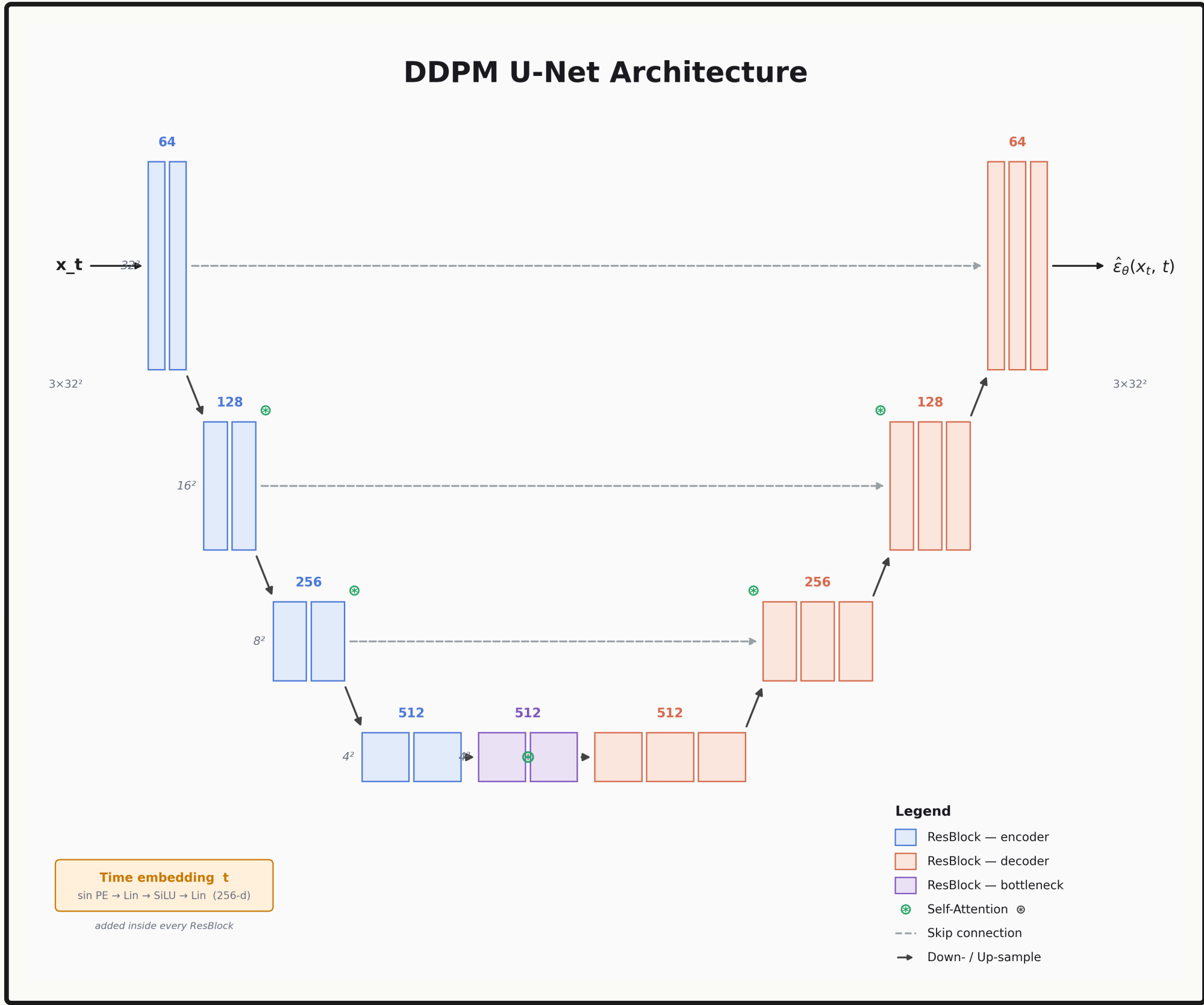


## 1. Motivation

- Diffusion = Gaussian noise — by construction.
- What if we strip the noise out entirely?
- Replace it with a deterministic blur.
- Does it still generate? **The paper says yes.**

## 2. Method

$$R_\theta(D(x, t), t) \approx x \quad \cdot \quad D = \text{blur}$$

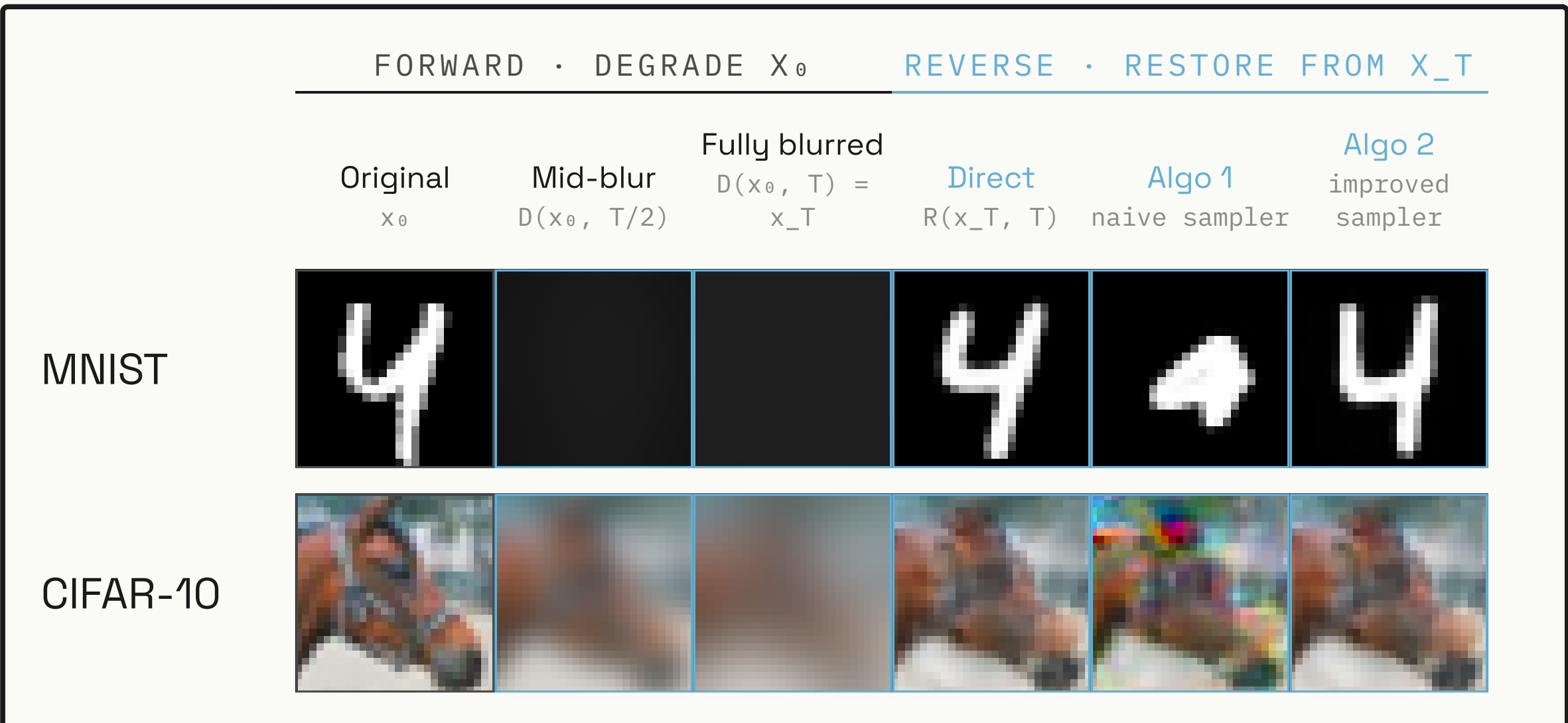


- U-Net: ResBlocks + attention + skip
- Sampling: Algorithm 2 (improved)
- Training: Adam, lr 2e-5

## 3. Datasets

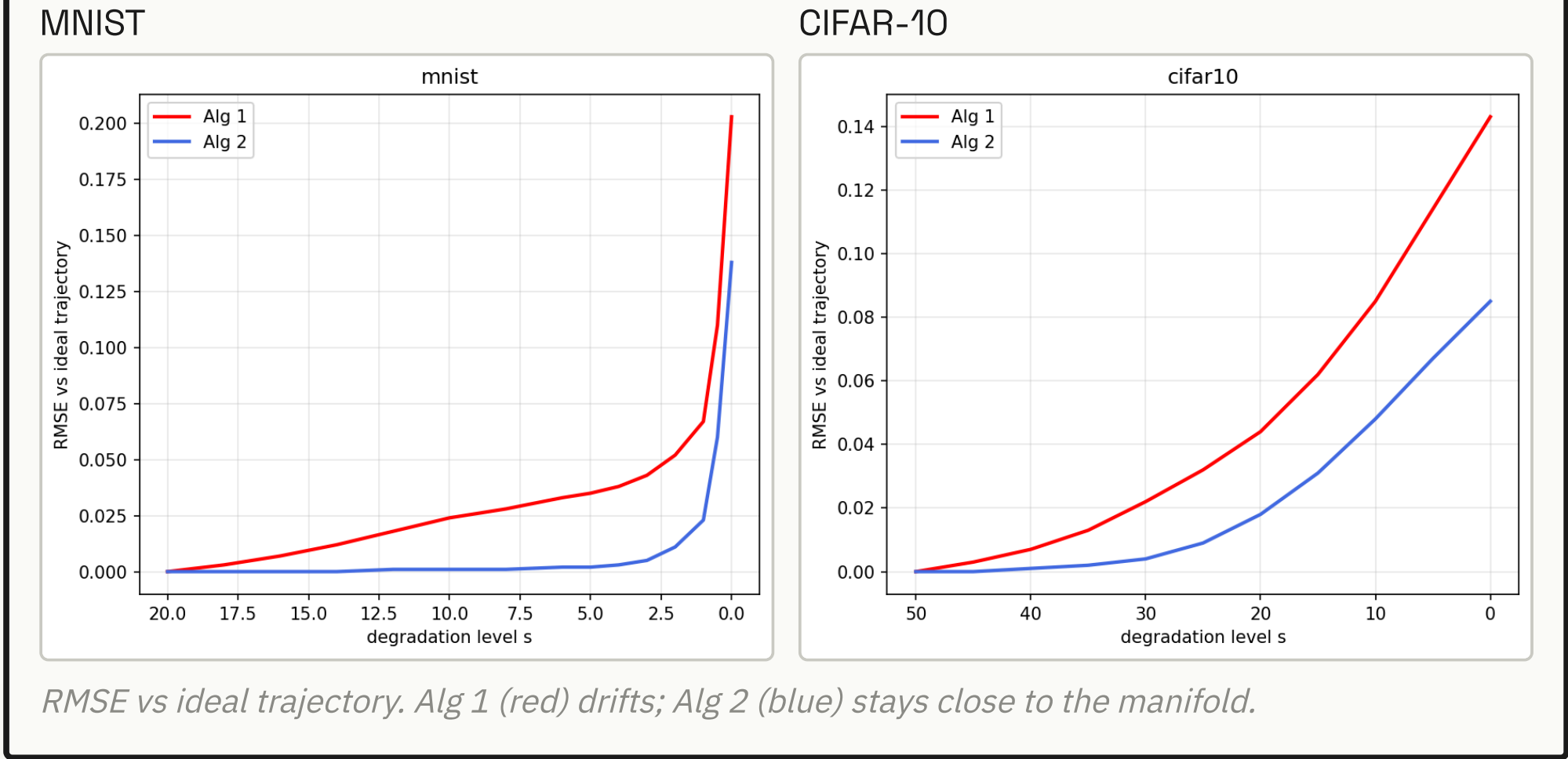
|                     |                       |
|---------------------|-----------------------|
| MNIST<br>28×28 gray | CIFAR-10<br>32×32 RGB |
|---------------------|-----------------------|

## ★ Forward freeze → reverse thaw



| ◆ Quantitative results FID / SSIM / RMSE · HELD-OUT TEST SET · LOWER IS BETTER |                   |        |        |        |        |        |        |
|--------------------------------------------------------------------------------|-------------------|--------|--------|--------|--------|--------|--------|
| Dataset                                                                        | Method            | Ours   |        |        | Paper  |        |        |
|                                                                                |                   | FID    | SSIM   | RMSE   | FID    | SSIM   | RMSE   |
| MNIST                                                                          | Degraded (x_T)    | 398.14 | 0.0141 | 0.2866 | 438.59 | 0.2870 | 0.2870 |
|                                                                                | Direct R(x_T, T)  | 27.04  | 0.8458 | 0.1358 | 5.10   | 0.7570 | 0.1420 |
|                                                                                | Alg 1 naive iter. | 41.16  | 0.6234 | 0.2374 | —      | —      | —      |
|                                                                                | Alg 2 improved    | 22.69  | 0.7537 | 0.1780 | 4.69   | 0.7180 | 0.1540 |
| CIFAR-10                                                                       | Degraded (x_T)    | 313.65 | 0.3562 | 0.1359 | 298.60 | 0.3150 | 0.1360 |
|                                                                                | Direct R(x_T, T)  | 58.33  | 0.6736 | 0.0810 | 83.69  | 0.7750 | 0.0710 |
|                                                                                | Alg 1 naive iter. | 96.42  | 0.5188 | 0.1205 | —      | —      | —      |
|                                                                                | Alg 2 improved    | 55.75  | 0.6634 | 0.0826 | 80.08  | 0.7730 | 0.0750 |

## 4. Results



### QUANTITATIVE DRIFT / STABILITY

- Per-step drift confirms the compounding error theory of §3.3.
- Alg 1 accumulates restoration error; Alg 2's self-correcting update keeps drift ≈ 0.
- Alg 2 < Alg 1 at every degradation level on both datasets.

## 5. Further experiments / exploration

- Random color removed at each timestep
- Focused gaussian blur around brighter points

